INDUSTRIAL SAFETY MONITORING HELMET

Arduino Source Code

Platform: Arduino IDE | Language: Embedded C++

A.1 Full Source Code

The program acquires temperature and humidity readings from the DHT11 sensor and an analog reading from the MQ-2 sensor, then displays the values through the Arduino IDE Serial Monitor.

```
#include <DHT.h>

#define DHTPIN 2
#define DHTTYPE DHT11
#define MQ2_PIN A0

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(9600);
  dht.begin();

  Serial.println("Smart Safety Helmet");
  Serial.println("System Started");
  Serial.println("------------------------");
}

void loop() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();
  int gasValue = analogRead(MQ2_PIN);

  if (isnan(humidity) || isnan(temperature)) {
    Serial.println("DHT11 Read Failed!");
  } else {
    Serial.print("Temperature: ");
    Serial.print(temperature);
    Serial.println(" C");

    Serial.print("Humidity: ");
    Serial.print(humidity);
    Serial.println(" %");
  }

  Serial.print("Gas Sensor Value: ");
  Serial.println(gasValue);
  Serial.println("------------------------");

  delay(2000);
}
```